

Objective:

This course is an introduction to the emerging field of Quantum Computers. Students are expected to know Quantum Mechanics but it is not mandatory. The objective of this course is to introduce the students to a different kind of logical system, namely, the quantum logic, based on the superposition of the states 0 and 1. The phenomenon of entangled states and its use in controlled operations are introduced. With this basic tools from the physics of Quantum Mechanics the course intends to prepare the students to understand some of the important algorithms in quantum computation, quantum Information and Cryptography.

Prerequisite: Quantum Mechanics, Linear Algebra.

1. Introduction to Quantum Mechanics

- (i) Postulates of Quantum Mechanics.
- (ii) Measurement in Quantum Mechanics.
- (iii) The bra-ket and density operator formalism.
- (iv) Composite (Multi qubit) System.

2. Introduction to Computer Science.

- (i) Turing Machine.
- (ii) Circuit model of computation.
- (iii) Measures of complexity.
- (iv) Energy for computation.
- (v) Reversible gates.

3. Quantum Circuits.

- (i) Qubits, Quantum Gates.
- (ii) Controlled Operations.
- (iii) Unitarity of gates.
- (iv) Relation to reversibility.

4. Quantum Algorithms.

- (i) Quantum Fourier Transforms.
- (ii) Order finding and factoring.
- (iii) Quantum Search Algorithms.
- (iv) Quantum Cryptography.

5. Quantum Noise

- (i) Quantum operations.
- (ii) Quantum Error Correction

Outcome:

The students learn the idea of qubits which implements parallel processing in quantum mechanics. The introduction of gates as Unitary operators acting on qubits help them to look at quantum computers as intrinsically reversible machines. The student learns a completely new algebra of logical system based on subspaces of a vector space as opposed to the boolean algebra based on subsets of a set. Controlled gates provides a concrete application of the complex phenomena of entanglement and measurement in quantum mechanics. The students learn the Deutsch's Algorithm, the search algorithm and algorithms based on Quantum Fourier Transform. This includes Shor's probabilistic algorithm for factoring an integer. The students learn the interesting application of generating a private key between parties using quantum phenomenon, rather than transmitting a predetermined key, in Quantum Cryptography. The course prepares the student to do further reading or get into research in the exciting field of Quantum Computation and Quantum Information.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
	X	X	X								X

Books:

1. Quantum Computation and Quantum Information
Michael A. Nielsen and Isaac L. Chuang
2. Modern Quantum Mechanics
J. J. Sakurai